Adaptive NetFlow IIoT Intrusion Detection With Deep Transfer Learning, Genetic Optimization, and Ensemble Methods for Network Management

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***Abstract*—The growing demands of Industry 5.0 necessitate resilient Internet of Things (IoT) networks, which are increasingly susceptible to sophisticated cyber threats. While advancements in intrusion detection systems (IDS) have improved attack detection, addressing the complexity of multi-class attack scenarios and managing minority threats remains challenging. This study proposes NFIIoT-DTL-IDS, an adaptive IoT IDS for smart network management using NetFlow and IIoT data, driven by deep transfer learning and enhanced with genetic algorithm (GA) optimization. Our framework leverages pre-trained CNN models to convert data into images, enabling effective classification. GA optimizes hyperparameters to improve model flexibility and performance, while a soft voting ensemble ensures robust aggregation of predictions. The proposed IDS achieves 100% classification accuracy among 5, 10, and 19 distinct attack classes of three IoT datasets, including two NetFlow datasets (NF-TON-IoTv2 and NF-BoT-IoTv2) and one industrial-based dataset (X-IIoTID), detecting threats such as DDoS, ransomware, and theft in common IoT cyberattacks, as well as MQTT subscription and crypto-ransomware attacks in industrial IoT scenarios. Additional experiments demonstrate the effectiveness of the proposed method, which exceeds the classification performance results of three baseline models, including LSTM, Transformer, and 3D CNN, by more than 37.2%, 0.25%, and 1.25%, respectively, in the maximum among the three datasets. Finally, experimental results demonstrate that NFIIoT-DTL-IDS outperforms recent state-of-the-art solutions in terms of multiclassification accuracy. This contribution advances adaptive management frameworks for IoT security, offering scalable and high-performance solutions for intrusion detection in modern network management.**

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I. Introduction

INDUSTRY 5.0, emphasizes the cooperation of humans and machines in industrial processes, aiming to produce more customized, adaptable, and sustainable production by fusing the capabilities of human labor with automated systems [1]. This paradigm emphasizes human-centric innovation, resilience, and sustainability, extending beyond the efficiency-driven goals of Industry 4.0 [2]. The Internet of Things (IoT) forms the backbone of Industry 5.0, enabling seamless data exchange among physical systems in industrial environments. However, this extensive connectivity introduces increasing security risks, including ransomware, malware, and denial-of-service attacks, often because of weak device configurations and insufficient encryption in common IoT systems. To mitigate these threats, robust security mechanisms are critical, particularly intrusion detection systems (IDS), which can monitor network activity in real time to detect and respond to anomalies [3]. IDS play a vital role in preserving the confidentiality, integrity, and availability of IoT systems by preventing unauthorized access and ensuring operational continuity. Despite advancement in using machine learning-based IDS for IoT security, two major challenges persist. First, the evolving and diverse nature of cyberattacks makes it difficult to detect and classify unknown threats using existing datasets [4]. Second, the heterogeneity of IoT data, spanning varied features and sources, complicates the construction of generalized models [5]. These challenges highlight the need for adaptive, intelligent detection systems capable of evolving alongside emerging threats and dynamic data environments.

A promising method for tackling this challenge is the application of deep transfer learning, which leverages pre-trained models to adapt quickly to new tasks with limited labeled data [6]. This approach helps address the scarcity of annotated IoT security data while maintaining high detection performance [7]. Unlike conventional machine learning methods, transfer learning leverages more effective classification performance of pre-trained models. By applying this capability to IoT security, deep learning-based transfer learning enables knowledge transfer from existing datasets to new or evolving environments, improving the detection of unknown attacks [6]. This approach remains effective even when the source and target data differ significantly, making it suitable for the heterogeneous and dynamic nature of IoT systems. As a result, transfer learning enhances the adaptability and robustness of intrusion detection in complex IoT settings.

Addressing the second challenge of building generalized IoT security models requires the use of datasets that accurately represent the diversity of real-world environments. This includes capturing a wide range of traffic patterns and up-to-date attack scenarios, such as common IoT threats like DDoS, ransomware, and data theft, that can cause significant disruption to IoT infrastructure [8]. Through incorporating these evolving threats, researchers can train more robust and comprehensive models capable of detecting varied attack types. To be specific, the unified features derived from NetFlow data can enhance generalization, as NetFlow captures rich, flow-level information across protocols and devices [9]. Furthermore, employing industrial IoT-based datasets can help to build a model that can be scaled into a real-world industrial IoT environment, which will serve as the foundation for industry 5.0 [10]. Thus, leveraging recent, industrial-scale datasets with consistent NetFlow-based features helps overcome the heterogeneity of IoT systems and supports the development of more effective security solutions.

Thus, this article introduces an effective and resilient IDS framework tailored for NetFlow-based IIoT environments, termed NFIIoT-DTL-IDS, which is grounded in deep transfer learning methodologies. Six pre-trained convolutional neural network (CNN) architectures, including Xception, MobileNet, MobileNetV2, Inception, DenseNet121, and EfficientNetB0, are utilized by the framework. The decision to use of the ideal hyperparameters has a significant impact on how well these pre-trained DL architectures work. Using a genetic algorithm (GA), the selected architectures are optimized and trained adaptively. The soft-voting ensemble technique is used to identify and combine the three most effective models after training and validation. The proposed framework's efficacy is then evaluated using two recent NetFlow-based datasets, NF-TON-IoTv2 and NF-BoT-IoTv2, as well as one industrial-based IoT security dataset. The study makes significant contributions in three main areas:

Data preprocessing and transformation: The proposed work applies a thorough preprocessing pipeline to NetFlow IoT and IIoT datasets, including NF-TON-IoTv2, NF-BoT-IoTv2, and X-IIoTID datasets. The pipeline includes preprocessing of missing values, duplicates, and irrelevant features and balancing by incorporating random under-sampling and oversampling techniques to address realistic and skewed distributions and conversion of original data into image representations to support subsequent deep learning.

Robust and adaptive NFIIoT-DTL-IDS Framework: A novel deep transfer learning-based intrusion detection system, NFIIoT-DTL-IDS, is proposed to effectively detect a wide range of cyberattacks in IoT and IIoT environments. The framework leverages six advanced pre-trained CNN architectures, each optimized using Genetic Algorithm (GA) for hyperparameters tuning. A lightweight soft-voting ensemble fuses the predictions of the top three models to enhance stability, accuracy, and generalization.

Extensive comparative performance evaluation: The proposed framework is evaluated on three IoT/IIoT datasets under multiclassification settings. Extensive comparisons are conducted with deep learning baselines including LSTM, Transformer, and 3D CNN architectures in this study. Furthermore, the framework's performance is benchmarked against several state-of-the-art approaches by recent literatures. Results consistently show superior performance of the proposed CNN-based ensemble in terms of classification accuracy among the attack classes, confirming its effectiveness across diverse real-world IoT environments.

The remainder of this work follows: Section II discusses related work and existing challenges in IDS for IoT. Section III presents the proposed NFIIoT-DTL-IDS framework, detailing each component of the methodology. Section IV shows the experimental environment and performance assessment. Section V provides results and discussion. Finally, Section VI concludes the study, summarizing its contributions and potential for future research.

II. Related Works

In artificial intelligence, deep transfer learning is an effective strategy that uses massive data sets with pre-trained models to improve target task performance with less labelled data [7]. Since most DL models require the arrangement of a huge quantity of labelled data, which is unachievable in many ML situations, in the domain of IoT intrusion detection, deep transfer learning demonstrates benefits over conventional machine learning and deep learning approaches. This section briefly covers the state-of-the-art machine learning, deep learning, transfer learning and approaches for IDS applications in IoT security.

Machine learning (ML) methods have shown promise in IoT IDS, but often require extensive manual feature engineering, particularly in dynamic environments. For example, correlation-based feature selection (CFS) has been used to reduce N-BaIoT's 115 features [11], and a wrapper-based method with bijective soft sets and CorrACC achieved high accuracy on BoT-IoT data [12]. Hybrid ML approaches, such as differential evolution combined with extreme learning machine, have also outperformed traditional combinations like CFS + SVM [13]. However, conventional ML approaches struggle with the scale and heterogeneity of IoT data. The evolution of deep learning addressed this by automatically extracting features from raw data, making it more viable in complicated IoT settings. These models excel at capturing intricate patterns and detecting subtle intrusions. For instance, autoencoder-based feature extraction improved detection rates and reduced error in industrial IoT IDS [14], [15]. Other approaches, such as dynamic line graph neural networks and MF-Net, have achieved high anomaly detection accuracy by modeling traffic as sequential data with varying frequencies [14], [16].

In addition, more deep learning models have been explored by recent studies. For instances, LSTM networks are widely explored for capturing temporal dependencies in sequential NetFlow traffic and have demonstrated improved intrusion detection performance in time-series scenarios [17], [18]. However, they are sensitive to long sequence dependencies and may underperform when handling high-dimensional, imbalanced data or requiring extensive tuning [19]. Transformer-based models, with their self-attention mechanisms, have shown promising results by capturing long-range relationships in traffic sequences more effectively [20], but their computational overhead and demand for large-scale data can be impractical for many real-world IoT deployments [21]. In addition, 3D CNNs offer another direction by jointly learning spatial and temporal features, yet they are often prone to overfitting and require significant computational resources [22]. Thus, these deep learning approaches require retraining or fine-tuning to maintain performance and scalability, highlighting the need for more scalable, adaptable, and resource-efficient models.

Deep transfer learning addresses these challenges by fine-tuning pre-trained models on generalized tasks to the specific nuances of IoT intrusion detection. It accelerates training and improves generalization across heterogeneous IoT scenarios, enhancing both accuracy and robustness. For example, [23] introduced a transfer learning approach designed for limited data availability, leveraging optimization methods like Lasso and MCP to improve learning efficiency. In [24], transfer learning was applied to intrusion detection for Internet-of-Cars systems using minimal labeled data. Their method achieved a 23% increase in accuracy on two AWID datasets, although these datasets lacked recent traffic patterns and were not specifically designed for IoT. Similarly, [25] proposed a federated transfer learning framework to detect RPL routing protocol intrusions in heterogeneous IoT networks. Using a custom RPL-IIoT dataset, their model achieved 85.52% accuracy, but it was limited to routing attacks and lacked benchmark validation.

Other studies have explored transforming traffic data into image representations to leverage pre-trained CNNs. In [26], DeepInsight was used to convert time-based features into 3D images, enabling the use of models like VGG19 and ResNet101. Their bi-level classification framework, coupled with classifiers such as SVM and RF, achieved 96% accuracy on Darknet traffic data. However, this work did not focus on IoT-specific traffic. In [27], TL-CNN-IDS combined transfer and ensemble learning to address class imbalance and limited training data. Tabular features were converted into image format for CNN processing, with model selection and optimization conducted using Tree-Structured Parzen Estimators. The final ensemble achieved 99.85% and 99.53% accuracy on CICIDS2017 and NSL-KDD, respectively, though both datasets are outdated and non-IoT-specific. Lastly, [6] developed a ResNet-based model using real-world traffic, achieving 87% accuracy in distinguishing attack from normal traffic even with limited labeled data. While effective, most of these approaches still lack validation on modern, domain-specific IoT datasets, limiting their practical generalizability.

Existing literature on Deep Transfer Learning (DTL) for intrusion detection in IoT networks has limitations that need to be addressed. Many strategies focus on outdated cybersecurity datasets and a limited number of cyberattacks, failing to represent the real IoT environment. To overcome this, our proposed approach utilizes the most recent IoT statistics to evaluate models, ensuring relevance to current IoT threats. Additionally, previous research lacks generalized models due to the variety of IoT datasets, hindering robustness in attack classification. We address this by using uniform NetFlow-based IoT datasets, enhancing the effectiveness of our approach in categorizing cyberattacks. Furthermore, previous investigations often use a single pre-trained model without optimization or integration, impacting accuracy and resilience. To this end, our method incorporates six pre-trained CNN architectures, optimizing hyperparameters with GA and employing a soft voting ensemble approach to enhance attack detection accuracy and robustness.

III. Framework

The proposed framework's overall research approach is presented in this section. It begins by outlining the datasets, data preprocessing, class balancing, data transformation, and data splitting. Next, it goes over the CNN pre-trained deep transfer learning method, hyperparameter optimization, and soft voting ensemble approach. Afterwards, performance comparison is conducted with the models including LSTM, Transformer, and 3D CNN, respectively, in order to validate the effectiveness of the proposed NFIIoT-DTL-IDS framework. Fig. 1 shows the proposed framework's block diagram and the performance comparison with the three benchmark models in brief.

A. Datasets

In this study, three up-to-date and diverse datasets were employed to comprehensively evaluate the proposed intrusion detection framework, reflecting both consumer and industrial IoT security scenarios. The NF-ToN-IoTv2 and NF-BoT-IoTv2 datasets are NetFlow-based and regenerated from the widely recognized TON-IoT [28] and BoT-IoT [29] datasets. Developed by the University of Queensland, these datasets simulate realistic, heterogeneous IoT environments with updated traffic and a broad range of modern cyberattacks, offering 10 and 5 distinct classes, respectively. Meanwhile, X-IIoTID represents an up-to-date Industrial IoT dataset structured according to the Industrial Internet Reference Architecture (IIRA) [10]. It includes 19 classes across over 820,000 instances and 59 features, capturing complex behaviors in smart factory and industrial control systems. Together, these three datasets form a comprehensive benchmark suite that ensures robust, real-world validation of the proposed NFIIoT-DTL-IDS framework across the spectrum of modern IoT and IIoT environments. The class distributions for these datasets are presented in Fig. 2

Out of the 16,940,496 total data flows in the NF-TON-IoTv2, 10,841,027 (63.99%) are attack samples and 6,099,469 (36.01%) are benign samples. Attack traffic in the dataset is classified into nine categories: ransomware, password, injection, DoS, DDoS, backdoor, scanning, and XSS. Of the 37,763,497 data flows in the NF-BoT-IoTv2, 37,628,460 (99.64%) are attack samples, while 135,037 (0.36%) are

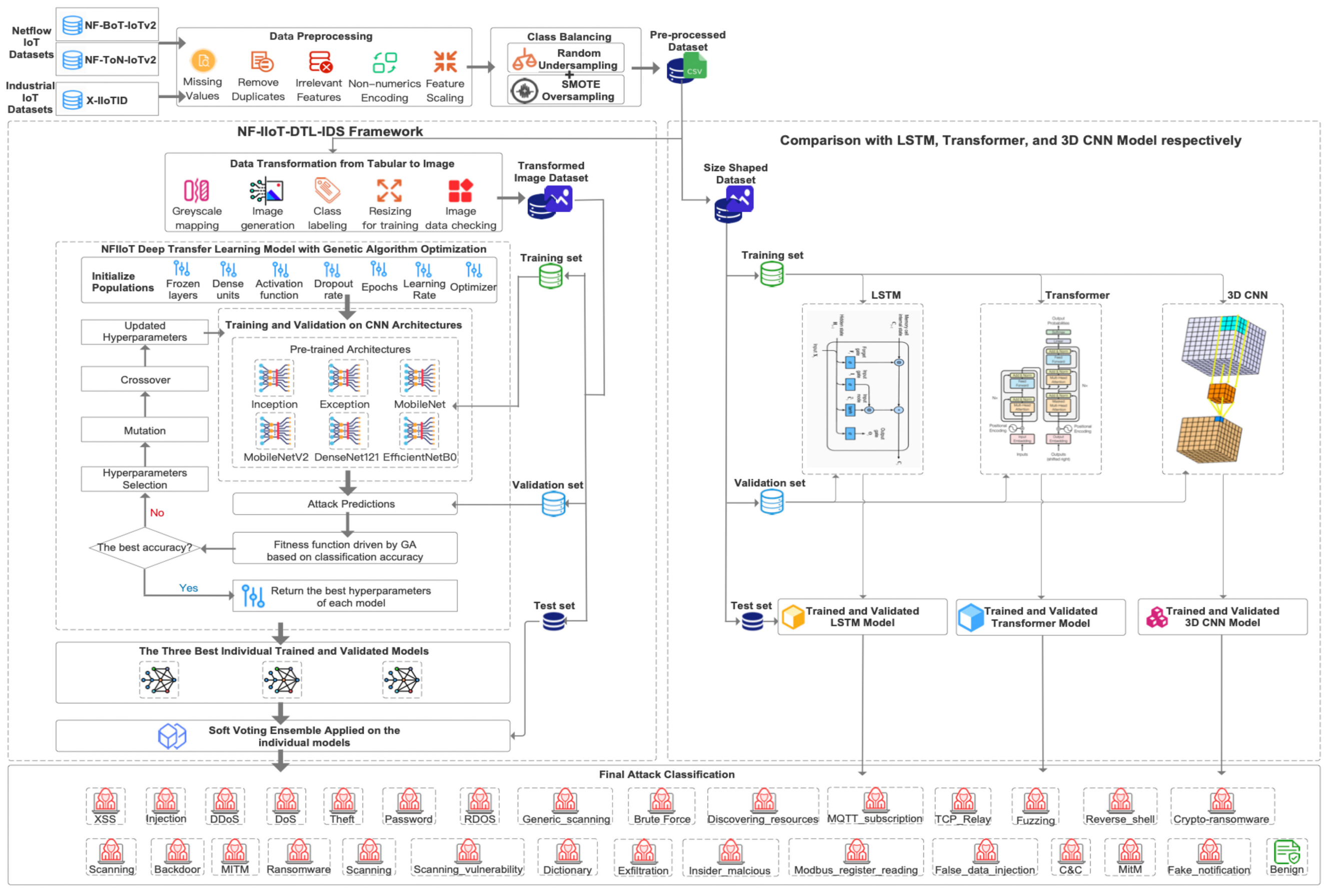

Fig. 1. Block diagram of NFIIoT-DTL-IDS framework.

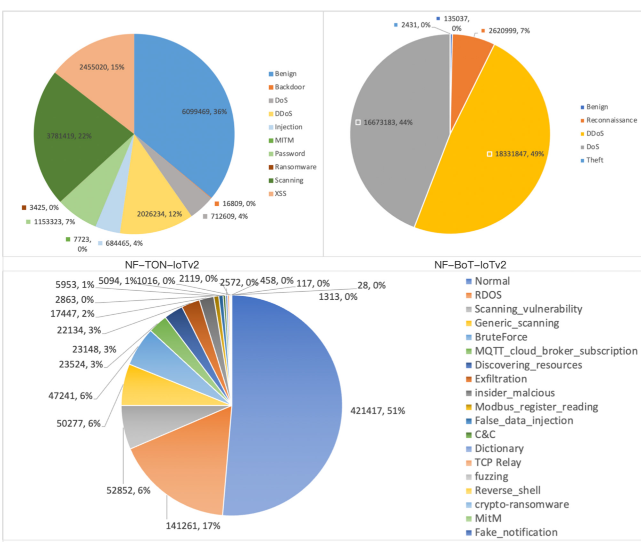

Fig. 2. Class distribution of the datasets.

benign samples. The dataset has 4 different types of attack traffic: Reconnaissance, DDoS, DoS, Theft. Each dataset traffic type has the same 4 flow identifier features, 8 flow features, and 31 extended flow features with the detailed description in [30]. In addition, the X-IIoTID dataset consists of 820,834 instances with 59 IIoT-specific features, offering a highly imbalanced distribution across 18 attack types, ranging from dominant classes like RDOS (17%) and Scanning Vulnerability (6%) to rare but critical threats such as Crypto-Ransomware (0.06%) and Fake Notification (<0.01%), thus reflecting the realistic and diverse nature of industrial IoT security scenarios.

B. Data Preprocessing

Optimal training of machine learning or deep learning models relies heavily on data preprocessing. The NF-TON-IoTv2 and NF-BoT-IoTv2 datasets consist of 43 features each, with two binary labels and 10 and 5 attack classes, respectively. After loading the datasets, we addressed missing values, finding none. Next, we removed 3,804,615 duplicate instances from NF-TON-IoTv2 (22.46% of dataset) and 7,343,411 from NF-BoT-IoTv2 (19.45% of dataset). Unnecessary identifier features (“IPV4_SRC_ADDR” and “IPV4_DST_ADDR”) were dropped, retaining “L4_SRC_PORT” and “L4_DST_PORT.” Post-drop, both datasets contained 13,135,881 and 30,420,086 instances, respectively, with 41 features. No numerical encoding was needed, except for the categorical feature “PROTOCOL,” which was dummy encoded into “PROTOCOL_ICMP,” “PROTOCOL_IGMP,” “PROTOCOL_TCP,” “PROTOCOL_UDP,” “PROTOCOL_IPv6_Frag,” and “PROTOCOL_IPv6_ICMP.” Finally, min-max normalization was applied, resulting in datasets of

13,135,881 rows with 48 columns and 30,420,086 rows with 47 columns. As for X-IIoTID dataset, three types of missing values (“NaN”, “-”, and “?”) were identified and replaced with –1 to indicate unavailability. Timestamp-related features were dropped as they serve no predictive value. A total of 4,260 duplicate rows were found and removed, leaving 816,574 unique instances. Two categorical features, “Protocol” and “Service”, were one-hot encoded, while remaining object-type features were converted to numerical float types. This transformation expanded the feature set from 65 to 79 features, except for the attack labels. Finally, min-max normalization was applied to scale the dataset.

C. Class Balancing

Imbalanced datasets pose significant challenges to classification models. The model may show biased behavior, have a low detection rate for minority classes, and have difficulty generalizing to new data due to the skewed distribution of class occurrences [31]. To address these limitations, techniques like over-sampling, under-sampling, and ensemble methods are commonly used to improve model performance on imbalanced datasets. In our study, we categorized classes into majority, minority, and highly minority ranks for the three preprocessed datasets, and in order to validate the robustness of the proposed NF-IIoT-DTL-IDS, diverse class balancing schemes are implemented on the three datasets, respectively.

For both NF-TON-IoTv2 and NF-BoT-IoTv2, considering there is a huge number of instances for the majority and minority classes in the two datasets, respectively, the random under-sampling technique is employed for the two ranked classes, while the oversampling technique using SMOTE generates synthetic instances for highly minority classes, except for “Benign” in NF-BoT-IoTv2, which has enough instances to be processed for subsequent data transformation. As for X-IIoTID, since there are no overwhelming instances for majority classes, which are kept as the original ones, thus, SMOTE is only applied for the highly minority classes including “Modbus_register_reading,” “Dictionary,” and “crypto-ransomware,” etc. Thus, diverse balancing schemes, including both undersampling and oversampling, are conducted on the two NetFlow IoT datasets, while the oversampling technique is employed on the IIoT dataset, which maintains the original distribution and ensures necessary balance among the classes while reducing computational complexity in IoT environments. Table I provides detailed information on the instances and class distributions.

D. Data Transformation

Within the proposed NFIIoT-DTL-IDS framework, we implemented a classification framework utilizing pre-trained CNN architectures. Typically, CNN models exhibit superior performance with image-based datasets. Given that the preprocessed NF-IoT datasets are structured akin to tabular data, the subsequent task involves transforming them into the format resembling that of an image-based dataset. Transforming tabular data into image data for input into pre-trained models involves converting the structured information into a visual

TABLE I

CLASS DISTRIBUTION OF THE DATA SETS

Attack type	Counts	Distribution	Balance Rank	Balancing Scheme	Counts	Distribution
NF-TON-IoTv2						
Benign	3601284	27.42%	Majority (>10%)	Random Under-sampling	180064	23.89%
Scanning	3002169	22.85%			150108	19.92%
XSS	2449955	18.65%			122497	16.26%
DDoS	1746590	13.30%			87329	11.59%
Password	993718	7.56%	Minority (1~10%)		49685	6.59%
Injection	660467	5.03%			33023	4.38%
DoS	654359	4.98%			32717	4.34%
Backdoor	16259	0.12%	Highly Minority (<1%)	Over-sampling (SMOTE)	32717	4.34%
MITM	7723	0.06%			32717	4.34%
Ransomware	3357	0.03%			32717	4.34%
Overall	13135881	100.00%			753574	100.00%
NF-BoT-IoTv2						
DDoS	18331847	48.54%	Majority (>10%)	Random Under-sampling	142802	29.78%
DoS	16673183	44.15%			136450	28.45%
Reconnaissance	2620999	6.94%	Minority (1~10%)		70890	14.78%
Benign	135037	0.36%	Highly Minority (<1%)		64718	13.49%
Theft	2431	0.01%		Over-sampling (SMOTE)	64718	13.49%
Overall	37763497	100.00%			479578	100.00%
X-IIoTID						
Normal	419275	51.35%	Majority (>10%)		419275	44.09%
RDOS	141261	17.30%		Keep the original classes	141261	14.85%
Scanning_vulnerability	52832	6.47%	Minority (1~10%)		52832	5.56%
Generic_scanning	50261	6.16%			50261	5.29%
BruteForce	47241	5.79%			47241	4.97%
MQTT_cloud_broker_subscription	23524	2.88%			23524	2.47%
Discovering_resources	23148	2.83%			23148	2.43%
Exfiltration	22134	2.71%			22134	2.33%
insider_malcious	15571	1.91%			15571	1.64%
Modbus_register_reading	5953	0.73%	Highly Minority (<1%)	Over-sampling (SMOTE)	15571	1.64%
False_data_injection	5094	0.62%			15571	1.64%
C&C	2863	0.35%			15571	1.64%
Dictionary	2475	0.30%			15571	1.64%
TCP_Relay	2119	0.26%			15571	1.64%
fuzzing	1204	0.15%			15571	1.64%
Reverse_shell	1016	0.12%			15571	1.64%
crypto-ransomware	458	0.06%			15571	1.64%
Overall	816429	100.00%			950957	100.00%

format [32]. Data points of features are mapped to grayscale picture points as the initial phase. Since each feature in the tabular data represents a pixel intensity value in a grayscale image, the assurance that every pixel intensity in a grayscale image falls within the acceptable range by mapping the feature values to the range [0, 255]. Pixel intensities of pictures in grayscale go from 0 (black) to 255 (white).

The second stage is creating an image. In our research, we do this by utilizing the RGB color model to convert the tabular data into a color image. Each pixel is represented by three values, involving red, green, and blue. These three-color channels combine to produce a wide spectrum of colors. Each channel's intensity value ranges from 0 to 255. The preprocessed NF-TON-IoTv2 dataset has 46 columns. Each block comprising 138 consecutive samples and 46 columns undergoes a transformation into a color image format with dimensions of $46\times46\times3$ (representing the three-dimensional height, width, and channels). Conversely, in the preprocessed NF-BoT-IoTv2 dataset, which consists of 45 columns, each block of 135 consecutive samples is transformed into a $45\times45\times3$ color picture [33], as depicted in Fig. 3. Similarly, for the X-IIoTID dataset, which contains 79 columns after

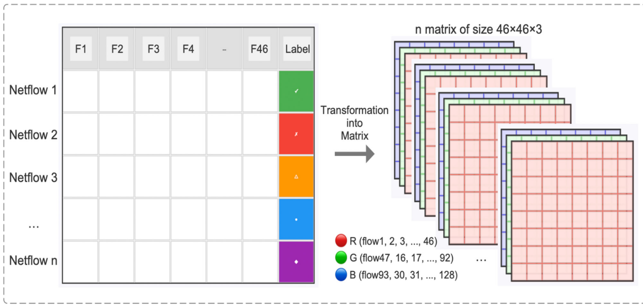

Fig. 3. NF-IIoT tabular data to matrix transformation.

preprocessing, each block of 237 consecutive samples is converted into a 79×79×3 color image, enabling visual pattern representation of IIoT traffic suitable for deep learning-based intrusion detection.

After transformation, images are labeled based on their original attack classes. Benign samples are labeled as "Benign", while DDoS-affected images are labeled "DDoS". This labeling extends to other attack classes. For NF-TON-IoTv2, images are sorted into 10 classes: Injection, DDoS, Benign, Password, XSS, DoS, Scanning, Backdoor, MITM, and Ransomware. NF-BoT-IoTv2 images are classified into 5 classes: DDoS, DoS, Reconnaissance, Benign, and Theft. Similarly, for X-IIoTID, images are categorized into 19 distinct classes, including RDOS, Brute Force, Fake Notification, and other representative industrial cyber threats. Resizing images to 224x224 pixels is crucial for CNN training, ensuring consistent input sizes, simplifying architecture design, and reducing memory requirements. Standardizing dimensions facilitates compatibility with pre-trained CNN models like VGG, Inception, or Xception. Resizing promotes better generalization across diverse datasets, focusing on extracting meaningful features. After resizing, image data is checked for each attack category, as shown in Fig. 4.

The Algorithm 1 outlines the main steps involved in preprocessing and generating images from tabular network flow data as described in the task. Each step is outlined with a placeholder function to represent the specific implementation details required for that step.

E. Data Splitting

Training, validation, and test sets are established from the transformed data. Cross-validation is often used to evaluate model performance, involving dividing the dataset into folds, training on subsets, and evaluating on the remainder. Due to data and computational constraints, a fixed validation set with sufficient epochs may be more practical. We monitor model performance on a consistent subset during training without frequent cross-validation. It's crucial to ensure the validation set adequately represents the data distribution [34]. Our study initially splits data into 80% training and 20% test sets. The original training set is then divided into new training and validation sets, each comprising 80% and 20%, respectively. Detailed statistics for the image dataset are presented in Table II.

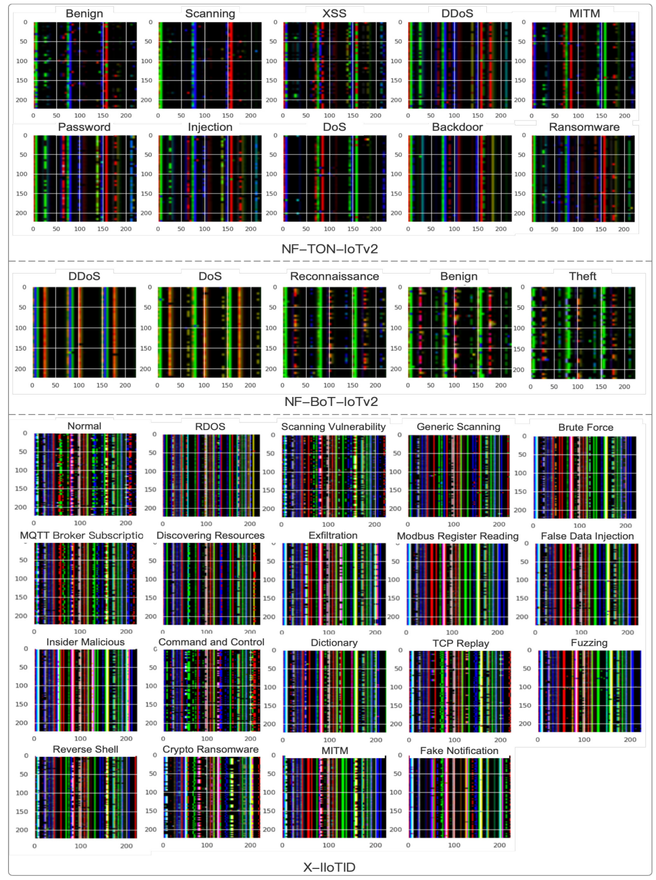

Fig. 4. Resized representative image data of the datasets.

Algorithm 1 Tabular-to-Image Transformation for NF-IIoT Data

Input: - Preprocessed tabular network flow data
Output: - Labeled color image dataset

Function *TransformTabularToImages*(*tabular_data*):
  Initialize *transformed_image_data* ← []
  For each *class_label* **in** unique(*tabular_data.labels*):
    class_data ← filter(tabular_data where label == *class_label*)
    For each block **in** *split_into_blocks*(*class_data, block_size*):
      image ← *convert_block_to_color_image*(*block*)
      resized_image ← resize(*image*, 224, 224)
    transformed_image_data.**append**({"image": *resized_image*, "label": *class_label* })
    representative ← *select_representative_image*(*transformed_image_data*, *class_label*)
      display(*representative*)
  Return *transformed_image_data*

F. CNN Transfer Learning Architectures

Convolutional Neural Networks (CNNs) are widely used deep learning models for pattern recognition, known for their ability to automatically extract hierarchical features through convolutional, pooling, and fully connected layers. This architecture enables strong generalization across tasks

TABLE II
THE STATISTICS OF THE TRANSFORMED NF-IIoT IMAGE DATA

Category	Train	Validation	Test	Total
NF-TON-IoTv2				
Benign	848	206	241	1295
Scanning	657	187	235	1079
XSS	565	138	178	881
DDoS	424	84	120	628
Password	238	46	73	357
Injection	139	49	49	237
DoS	580	123	160	235
Backdoor	70	18	28	116
MITM	31	11	13	55
Ransomware	12	7	5	24
Overall	3140	785	982	4907
NF-BoT-IoTv2				
Benign	292	72	111	475
DoS	1313	333	407	1003
Theft	10	2	5	17
DDoS	678	161	211	1050
Reconnaissance	346	84	91	521
Overall	1961	491	614	3066
X-IIoTID				
Normal	1127	282	352	352
RDOS	379	95	118	118
Scanning_vulnerability	142	35	44	44
Generic_scanning	134	34	42	42
BruteForce	126	32	39	39
MQTT_cloud_broker_ subscription	63	16	19	19
Discovering_resources	62	15	19	19
Exfiltration	59	15	18	18
insider_malcious	42	10	13	13
Modbus_register_reading	50	12	9	9
False_data_injection	47	12	8	8
C&C	41	10	10	10
Dictionary	40	10	10	10
TCP Relay	39	10	10	10
fuzzing	20	5	9	9
Reverse_shell	19	5	9	9
crypto-ransomware	18	4	8	8
MitM	17	4	8	8
Fake_notification	17	4	8	8
Overall	2442	610	753	3052

such as image processing and natural language understanding. Deep transfer learning builds on this capability by adapting pretrained models, like CNNs, to new but related tasks, especially where labeled data is scarce [35]. In this study, the proposed NFIIoT-DTL-IDS framework uses six CNN architectures, namely Xception, Inception, MobileNet, MobileNetV2, DenseNet121, and EfficientNetB0, which are commonly pre-trained on the ImageNet database [36], a large-scale visual database containing millions of labeled images across thousands of categories. The pre-training on ImageNet involves training these models on a vast array of images from diverse categories, helping the models learn general features and patterns present in various visual data.

To be specific, each CNN architecture brings unique strengths to the task of attack classification for transformed IoT image data [37]. The Inception architecture captures multi-scale features using parallel branches with 1×1, 3×3, and 5×5 convolutions and max pooling. Xception builds on this by using depthwise separable convolutions for improved efficiency and spatial feature extraction. MobileNet also employs depthwise separable convolutions to reduce complexity, making it suitable for resource-constrained IoT devices. MobileNetV2 enhances this with inverted residuals and shortcut connections for better accuracy at low cost. DenseNet introduces dense inter-layer connections to promote feature reuse and mitigate vanishing gradients. EfficientNet uses compound scaling to balance depth, width, and resolution, with EfficientNetB0 delivering strong performance under minimal computational load, which is ideal for IoT scenarios. The selection of diverse architectures strategically leverages their unique strengths to improve the model's attack classification accuracy in resource-limited IoT security settings. Consequently, individually optimizing the hyperparameters of these models and ensemble them within the NFIIoT-DTL-IDS framework offers a robust and comprehensive intrusion detection approach for IoT environments.

G. Hyperparameters Optimization Using Genetic Algorithm

Fine-tuning hyperparameters plays a pivotal role in optimizing and strengthening learning models by systematically adjusting configurable settings that influence the learning process but are not learned from the data [38]. This process significantly enhances a model's performance, generalization, and robustness by identifying optimal hyperparameter values. In our approach, we selectively unfreeze top layers of pre-trained architectures, allowing the model to adapt and refine its representations with new datasets while preserving knowledge from the source task. This strategy enables the model to better capture characteristics of the new task, ultimately improving performance in the target domain.

Common optimization algorithms for hyperparameter tuning include Grid search (GS), Random search (RS), Gradient-based algorithms (GO), Bayesian optimization (BO), and Metaheuristic algorithms like PSO and GA. Each method offers advantages and drawbacks. GS is simple to implement but time-consuming, especially with a wide range of hyperparameters, and may struggle with continuous parameters. RS efficiently explores a broad search space by randomly selecting values but may involve unnecessary evaluations and lacks consideration of previous results, reducing efficiency. Gradient-based algorithms (GO) are less common for hyperparameter optimization as they only work with continuous hyperparameters and are limited in finding local optima in non-convex issues rather than global ones [39]. Compared to GS and RS, Bayesian Optimization (BO) has a lower computing cost [40]. Particle Swarm Optimization (PSO), as one of the metaheuristic algorithms, excels in extensive parallelization and is well-suited for both continuous and conditional hyperparameter optimization [41]. Genetic Algorithms (GA), a type of metaheuristic, is complex but effective for optimizing diverse hyperparameters, supporting various types and excelling in large configuration spaces with the potential for near-optimal solutions even in a few iterations [42]. Considering multiple types of hyperparameters of complex CNN network-based algorithms in diverse contexts of IoT environments, GA can be the choice since it is suitable for exploring large solution spaces efficiently, and it is robust against local optima due to its stochastic nature. Moreover, it can handle both continuous and discrete optimization problems.

In this study, GA was employed to obtain the best hyperparameters for selected pre-trained models effectively. The

Algorithm 2 Optimized Hyperparameters Driven by GA

Input:
pop_size: Population size
num_generations: Number of generations
mutation_rate: Mutation rate
pretrained_models: Selected pre-trained CNN architectures
training_data, validation_data: Datasets for training and validation
Output:
best_hyperparameters, best_fitness_score

```
Define hyperparameter_space ← {
    'frozen_layers', 'dense_units', 'activation_function',
    'dropout_rate', 'optimizer', 'learning_rate', 'epochs'
}
Initialize population ← initialize_population(pop_size, hyperparameter_space)
For generation in range(num_generations) do:
    Evaluate fitness_scores ← [
        evaluate_fitness(individual, pretrained_models, training_data, validation_data)
        for individual in population ]
    Select parents ← select_parents(population, fitness_scores)
    Generate offspring ← crossover_and_mutate(parents, mutation_rate, hyperparameter_space)
    Update population ← replace_population(population, offspring, fitness_scores)*
End For
Select best_individual ← select_best(population, fitness_scores)
Decode best_hyperparameters ← decode_individual(best_individual)
Evaluate best_fitness_score ← evaluate_fitness(best_individual, pretrained_models, training_data, validation_data)
Return best_hyperparameters, best_fitness_score
```

process involves initializing a population of potential solutions with defined hyperparameters, followed by fitness evaluation through training and validation on a dataset. Selection methods like tournament or roulette wheel selection are used to create a mating pool from individuals with higher fitness scores. Crossover operations exchange genetic material to generate offspring, while mutation introduces random alterations to sustain genetic diversity. This iterative process continues until termination conditions are met when reaching a maximum number of generations. The model with the highest fitness score undergoes further training and validation using optimized hyperparameters. Finally, the top 3 models demonstrating the highest performance scores are selected for ensemble operations, aiming for optimal classification performance. Detailed steps of the entire GA-based HPO procedure are delineated in Algorithm 2.

H. Soft Voting Ensemble Method

Ensemble methods in classification aim to enhance overall performance by combining predictions from multiple base classifiers [43]. Voting is a simple and computationally inexpensive ensemble method that combines predictions without training a meta-model. It mitigates the risk of overfitting compared to methods like stacking or boosting. Additionally, voting is versatile and can be applied to any base classifier, making it adaptable to various tasks and datasets in resource-constrained IoT environments. In classification ensemble methods, hard voting involves each classifier voting for a class, with the majority winning, while soft voting considers the anticipated probabilities from classifiers, potentially improving decision-making by leveraging more information. The process of soft voting ensemble in this study is elaborated in Algorithm 3.

Algorithm 3 Soft Voting Ensemble Method

Input:
input_data: Test data
base_classifiers: List of trained classifiers
Output:
final_predictions: Predicted class labels for the input data

```
Function soft_voting(input_data, base_classifiers):
    Initialize combined_probabilities ← zeros(shape=(num_samples, num_classes))
    For clf in base_classifiers do:
        probs ← clf.predict_proba(input_data)
        combined_probabilities ← combined_probabilities + probs
    Normalize combined_probabilities ← combined_probabilities / len(base_classifiers)
    final_predictions ← argmax(combined_probabilities, axis=1)
    Return final_predictions
```

I. Comparison With LSTM, Transformer, and 3D CNN Models

To validate the robustness and generalizability of the proposed image-based intrusion detection approach, we extend the methodology using three sequence-learning baselines: LSTM, Transformer, and 3D CNN models. Unlike the image-based models that rely on visual pattern recognition, these baselines operate directly on preprocessed tabular data that is structured into time-series sequences. For each dataset, the tabular features are segmented into fixed-length sequences, ensuring that each sequence consists of samples from the same class to preserve semantic consistency during training.

The sequence inputs are normalized, and the class labels are one-hot encoded. The resulting tensors follow the format (samples, timesteps, features), suitable for both LSTM and Transformer models. For the 3D CNN model, the data is further reshaped into 5D tensors by adding two singleton spatial dimensions, enabling spatial-temporal feature extraction through 3D convolutional operations. Thus, these tensors are further reshaped to (samples, timesteps, features, 1, 1), allowing the convolutional layers to learn spatial-temporal patterns across both dimensions.

To ensure consistency and fair comparison across all models, the same data splitting scheme is applied as the same

TABLE III
INPUT DATA SHAPES FOR THE MODELS AND DATASET AFTER SPLITTING

Model	Train set	Validation set	Test set
NF-TON-IoTv2			
LSTM / Transformer	(10,481, 46, 46)	(2,621, 46, 46)	(3,276, 46, 46)
3D CNN	(10,481, 46, 46, 1, 1)	(2,621, 46, 46, 1, 1)	(3,276, 46, 46, 1, 1)
NF-BoT-IoTv2			
LSTM / Transformer	(6,819, 45, 45)	(1,705, 45, 45)	(2,132, 45, 45)
3D CNN	(6,819, 45, 45, 1, 1)	(1,705, 45, 45, 1, 1)	(2,132, 45, 45, 1, 1)
X-IIoTID			
LSTM / Transformer	(11,608, 45, 79)	(2,902, 45, 79)	(3,628, 45, 79)
3D CNN	(11,608, 45, 79, 1, 1)	(2,902, 45, 79, 1, 1)	(3,628, 45, 79, 1, 1)

scheme applied in Section III-E, Data splitting for image-based data. This stratified 3-way split ensures that the image-based CNN models and the sequence-learning models (LSTM, Transformer, 3D CNN) are trained and evaluated on the exact same data distributions. The detailed dataset-specific settings, including sequence length, feature dimensions, and class distribution, are detailed in Table III. The hyperparameters and model configurations are kept consistent across experiments and are presented in the subsequent section.

IV. Experimental Environment and Performance Assessment

This section outlines the experimental setup and assesses the proposed scheme's performance. It commences with a delineation of the performance evaluation metrics, the platform employed for model training, the hyperparameters utilized, and the settings of the comparison models.

A. Experimental Environment

The NFIIoT-DTL-IDS framework is deployed and evaluated for performance on the Kaggle platform, leveraging Keras libraries. Kaggle provides access to a Nvidia Tesla P100 GPU equipped with 16 GB of graphic card memory and 19.5 GB of RAM, ensuring seamless execution of neural network-based algorithms. Initially, the raw network flow-based IoT data is collected and transformed into image datasets along with corresponding labels on the Kaggle platform. Subsequently, six pre-trained CNN models are integrated into the Keras applications with predefined and optimized hyperparameters. In addition, to establish comprehensive performance benchmarks, we further implement and evaluate three additional deep learning architectures: LSTM, Transformer, and 3D CNN, which are widely used in time-series and spatial-temporal modeling tasks. These models are trained on the same datasets and settings to ensure fair comparison. All best-performing models, including CNN-based and comparison models, are archived on Kaggle drive for final inference and prediction of unknown test data.

B. Performance Assessment Metrics

Several performance metrics, including as accuracy, precision, recall, F1 score, Cohen's Kappa (CK) Score, Matthews Correlation Coefficient (MCC), and confusion matrices, are used to evaluate the performance of the proposed NFIIoT-DTL-IDS system. The following formulas are used to compute these measures. A comprehensive indicator of the model's correctness, accuracy is the ratio of properly predicted occurrences to the total instances; nonetheless, it might not be the best choice for datasets that are unbalanced. Precision emphasizes the accuracy of positive predictions and reduces false positives by expressing it as the ratio of accurately predicted positive observations to the total expected positives. Recall highlights the model's capacity to find all pertinent cases and minimize false negatives by expressing it as the ratio of accurately predicted positive observations to all actual positives.

The performance metrics represented by equations (1) through (6) are displayed below.

$$Accracy = \frac{(TP + TN)}{(TP + FN + FP + TN)} \tag{1}$$

$$Precision = \frac{TP}{(TP + FP)} \tag{2}$$

$$Recall = \frac{TP}{(TP + FN)} \tag{3}$$

$$F1 - Score = \frac{2 * Precision * Recall}{Presicion + Recall} \tag{4}$$

$$Cohen\ Kappa\ score = \frac{Observed\ Agreement - Expected\ Agreement}{1 - \ Expected\ Agreement} \tag{5}$$

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{6}$$

The F1 score, derived as the harmonic mean of precision and recall, offers a balanced assessment, particularly valuable in scenarios of disparate class distributions within IoT data. Cohen's Kappa (CK) score serves as a statistical indicator evaluating agreement between two raters, surpassing chance expectations, notably advantageous when confronting imbalanced datasets. The Matthews Correlation Coefficient (MCC) emerges as a performance metric gauging classification model quality. It considers TP, TN, FP, and FN, yielding a nuanced assessment of model performance, particularly crucial in imbalanced class contexts.

C. Hyperparameters Schemes

To optimize the performance of neural network-based techniques, hyperparameters in deep transfer learning models must be carefully tuned. Previous research highlighted the significant impact of network type, depth, fully connected layer neuron count, optimizer selection, and learning rate adjustment on classification outcomes [44]. In this study, we employ Genetic Algorithm (GA) for Hyperparameter Optimization (HPO) across seven key parameters including, frozen layers, dense units, activation functions, dropout rates, optimizers, learning rates, and epochs within our NFIIoT-DTL-IDS framework. The optimized hyperparameters are then used to train all CNN models, with performance evaluated based on classification accuracy. Two configurations are compared:

TABLE IV
ORIGINAL HYPERPARAMETER SCHEMES OF PRE-TRAINED MODELS

Model	Hyperparameter						
	Frozen layers	Dense neurons	Activation function	Drop out Rate	Optimizer	Learning Rate	Epochs
NF-TON-IoTv2 / NF-BoT-IoTv2/X-IIoTID							
Xception	120	256	Relu	0.5	Adam	0.050	25
InceptionV3	100	256	Relu	0.5	Adam	0.050	25
MobileNet	50	256	Relu	0.2	Adam	0.050	25
MobileNetV2	100	256	Relu	0.2	Adam	0.003	25
DenseNet121	100	128	Relu	0.5	Adam	0.009	25
EfficientNetB0	200	256	Relu	0.2	Adam	0.003	25

TABLE V
OPTIMIZED HYPERPARAMETERS FOR THE PRE-TRAINED CNN MODELS

Model	Hyperparameter						
	Frozen layers	Dense neurons	Activation function	Drop out Rate	Optimizer	Learning Rate	Epochs
NF-TON-IoTv2							
Xception	118	128	Elu	0.4	adamax	0.0097	18
InceptionV3	112	128	Elu	0.3	adagrad	0.0012	19
MobileNet	136	256	Elu	0.5	adamax	0.0051	19
MobileNetV2	117	128	Relu	0.3	adagrad	0.0047	20
DenseNet121	104	256	Elu	0.6	adagrad	0.0071	17
EfficientNetB0	119	256	Elu	0.3	nadam	0.0097	20
NF-BoT-IoTv2							
Xception	118	128	Elu	0.4	adamax	0.0097	18
InceptionV3	112	128	Elu	0.3	adagrad	0.0012	19
MobileNet	136	256	Elu	0.5	adamax	0.0051	19
MobileNetV2	117	128	Relu	0.3	adagrad	0.0047	20
DenseNet121	104	256	Elu	0.6	adagrad	0.0071	17
EfficientNetB0	119	256	Elu	0.3	nadam	0.0097	20
X-IIoTID							
Xception	147	256	Elu	0.6	adagrad	0.0077	31
InceptionV3	148	256	Relu	0.4	ddamax	0.0069	25
MobileNet	113	256	Selu	0.1	nadam	0.0096	27
MobileNetV2	146	256	Selu	0.2	adamax	0.0022	18
DenseNet121	143	256	Relu	0.1	adamax	0.0096	22
EfficientNetB0	116	256	Elu	0.2	adamax	0.0084	14

default settings and those optimized via the HPO algorithm. One configuration represents the default settings for all base learning models, as outlined in Table IV, while the other is optimized through the HPO algorithm and presented in Table V.

D. Settings of the Comparison Models

To ensure fair and reproducible evaluation against the proposed CNN image-based models, the three baseline models are configured with consistent training hyperparameters where applicable, including the use of the Adam optimizer with a learning rate of 0.02, categorical cross-entropy loss, and 25 training epochs. The LSTM model utilizes two stacked LSTM layers with 128 units each, followed by a dense layer of 256 units. The Transformer model employs a 64-dimensional embedding, a 4-head attention mechanism, and a feed-forward network with 128 units. The 3D CNN architecture consists of two convolutional blocks with 32 and 64 filters respectively, followed by a fully connected layer with 256 units. Dropout regularization is applied to mitigate overfitting. All layers are fully trainable, and no frozen layers are used. These configurations are summarized in Table VI to facilitate reproducibility and performance comparison.

TABLE VI
KEY HYPERPARAMETERS AND SETTINGS OF COMPARISON MODELS

Model	Hyperparameter						
	Frozen layers	Dense units	Activation function	Dropout	Optimizer	Learning Rate	Epoch
NF-TON-IoTv2/ NF-BoT-IoTv2/ X-IIoTID							
LSTM	None	LSTM: 128 × 2 Dense: 256	ReLU / Softmax	0.2	Adam	0.02	25
Transformer	None	Embed: 64 FFN: 128	ReLU / Softmax	0.2	Adam	0.02	25
3D CNN	None	Conv3D: 32, 64 Dense: 256	ReLU / Softmax	0.2	Adam	0.02	25

V. RESULTS AND DISCUSSION

In this section, two sets of results will be presented and evaluated: one will assess the effectiveness of the proposed techniques within the NFIIoT-DTL-IDS framework, and the other will evaluate the effectiveness of the proposed framework against the three baseline models through experiments and comparisons with state-of-the-art works, finally followed by discussion.

A. Performance Evaluation

To evaluate transfer learning models and hyperparameter optimization, three stages of experimentation and analysis are conducted using NF-TON-IoTv2 and NF-BoT-IoTv2 datasets.

The first stage assesses individual model performance. Prediction results for all models without hyperparameter optimization (HPO) are presented in Table VII. In NF-TON-IoTv2, InceptionV3 achieves the highest accuracy (99.80%) and effectively classifies nine out of ten attack types, while Exception and DenseNet121 also perform well. However, MobileNetV2 and EfficientNetB0 show lower effectiveness. In NF-BoT-IoTv2, Xception, MobileNet, and DenseNet121 demonstrate high accuracy (99.35%) and effectively classify four out of five attack types. In NF-TON-IoTv2, InceptionV3 outperforms other models with 99.80% accuracy, correctly classifying nine out of ten attack types. Exception and DenseNet121 also show decent performance, while MobileNetV2 and EfficientNetB0 perform less effectively. In NF-BoT-IoTv2, Xception, MobileNet, and DenseNet121 achieve high accuracy (99.35%) and effectively classify four out of five attack types, except for Theft. InceptionV3 and MobileNetV2 exhibit moderate performance, while EfficientNetB0 performs poorly across all metrics.

The subsequent phase involves the selection and aggregation of outcomes from several top-performing models. Three models exhibiting the highest attack detection accuracy are chosen, including Xception, InceptionV3, and DenseNet121, for both datasets. Figures 7 and 8 illustrate a comparison of these outstanding models in the non-HPO scenario, with their respective bar graphs depicted on the left side. The outputs of these models are merged utilizing the proposed Soft Voting Ensemble methodology, which proves advantageous in preserving model accuracy and enhancing robustness of the models. The final ensemble model for NF-TON-IoTv2

TABLE VII
PERFORMANCE METRICS OF NON-HPO MODELS ON TEST DATA

Model	Performance Metrics					
	Accuracy	Precision	Recall	F1 score	CK score	MCC
NF-TON-IoTv2						
Xception	0.9837	0.9721	0.9837	0.9772	0.9802	0.9804
InceptionV3	**0.9980**	**0.9981**	**0.9980**	**0.9977**	**0.9975**	**0.9975**
MobileNet	0.9409	0.8961	0.9409	0.9159	0.9276	0.9304
MobileNetV2	0.8065	0.7828	0.8065	0.7701	0.7621	0.7732
DenseNet121	0.9847	0.9758	0.9847	0.9788	0.9814	0.9816
EfficientNetB0	0.3045	0.1641	0.3045	0.2126	0.0814	0.0973
NF-BoT-IoTv2						
Xception	**0.9935**	**0.9873**	**0.9935**	**0.9903**	**0.9909**	**0.9910**
InceptionV3	0.6759	0.5105	0.6759	0.5681	0.5086	0.5869
MobileNet	**0.9935**	**0.9873**	**0.9935**	**0.9903**	**0.9909**	**0.9910**
MobileNetV2	0.6498	0.5275	0.6498	0.5812	0.4992	0.5193
DenseNet121	0.9934	0.9872	0.9934	0.9903	0.9909	0.9910
EfficientNetB0	0.3192	0.1019	0.3192	0.1545	0.0000	0.0000
X-IIoTID						
Xception	**0.9987**	**0.9988**	**0.9987**	**0.9987**	**0.9982**	**0.9982**
InceptionV3	0.9734	0.9608	0.9734	0.9655	0.9643	0.9646
MobileNet	0.9880	0.9816	0.9880	0.9840	0.9839	0.9841
MobileNetV2	0.5671	0.4837	0.5671	0.4776	0.3611	0.4019
DenseNet121	0.9774	0.9649	0.9774	0.9697	0.9697	0.9700
EfficientNetB0	0.8035	0.6906	0.8035	0.7375	0.7231	0.7316

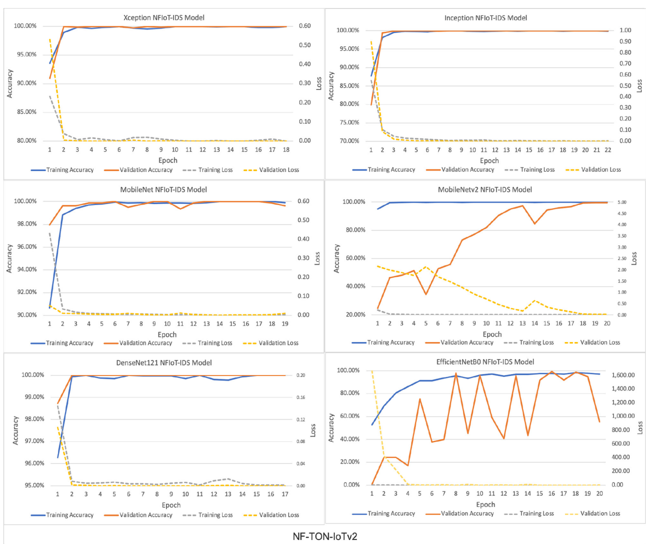

Fig. 5. Training and validation results of optimized pre-trained models on NF-TON-IoTv2.

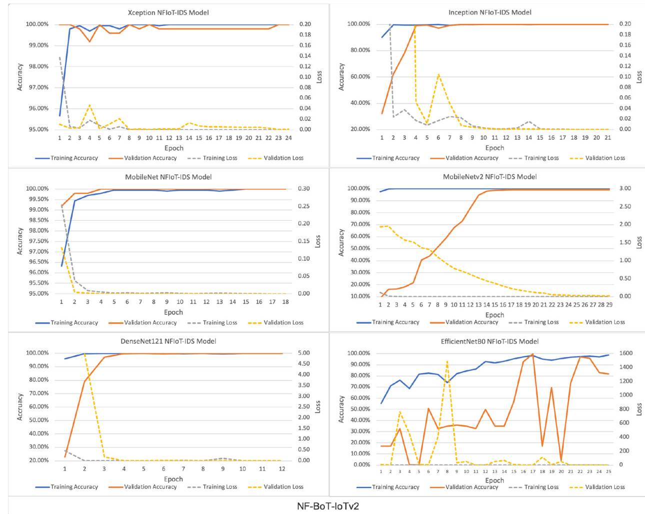

Fig. 6. Training and validation results of optimized pre-trained models on NF-BoT-IoTv2.

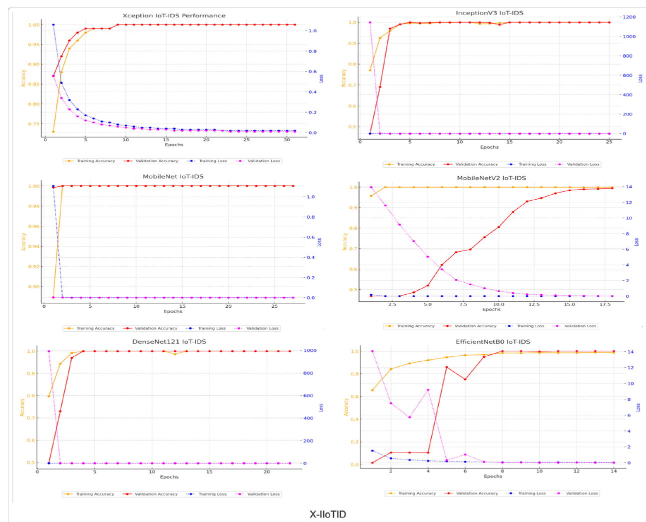

Fig. 7. Training and validation results of optimized pre-trained models on X-IIoTID.

achieves an accuracy of 98.47%, while for NF-BoT-IoTv2, it reaches 99.34%. Additionally, the final IDS for NF-TON-IoTv2 attains precision, recall, F1, CK score, and MCC values of 97.04%, 98.47%, 97.73%, 98.14%, and 98.16%, respectively. For NF-BoT-IoTv2, these metrics are recorded as 98.73%, 99.35%, 99.03%, 99.09%, and 99.10%, respectively. Figs. 9 and 10 show the confusion matrices of the final ensemble model in the non-optimized scenario on the left side.

In the second phase, the performance of individual transfer learning models with HPO is presented and evaluated, moreover, the impact of models optimized by HPO is additionally assessed, and the top three performing models are chosen and amalgamated for the Soft Voting ensemble model. All base learning models are trained using the optimized hyperparameters determined by GA, as outlined in Table V. In each case, the training and validation accuracies along with their respective losses for the optimized scenario are visualized and depicted in Fig. 5, 6 and 7 for NF-TON-IoTv2, NF-BoT-IoTv2, and X-IIoTID, respectively.

In NF-TON-IoTv2, the four figures of Xeption, Inception, MobileNet, and DenseNet121 show that the optimized accuracy for both training and validation approaches converged, while the loss of both training and validation approaches minimal value starting from the 3rd epoch. The four models show a consistent pattern between training and validation accuracy, with the corresponding training loss and validation loss. MobileNetv2's validation accuracy curve begins to converge around the 18th epoch, while the validation loss approaches the training loss by the same epoch. However, the EfficientB0 model differs from the other five models in that although the training accuracy increases, the validation accuracy fluctuates until the optimized final epoch.

In NF-BoT-IoTv2, the four figures of Xeption, Inception, MobileNet, and DenseNet121 show nearly the same pattern of those in the aforehand dataset. the four models

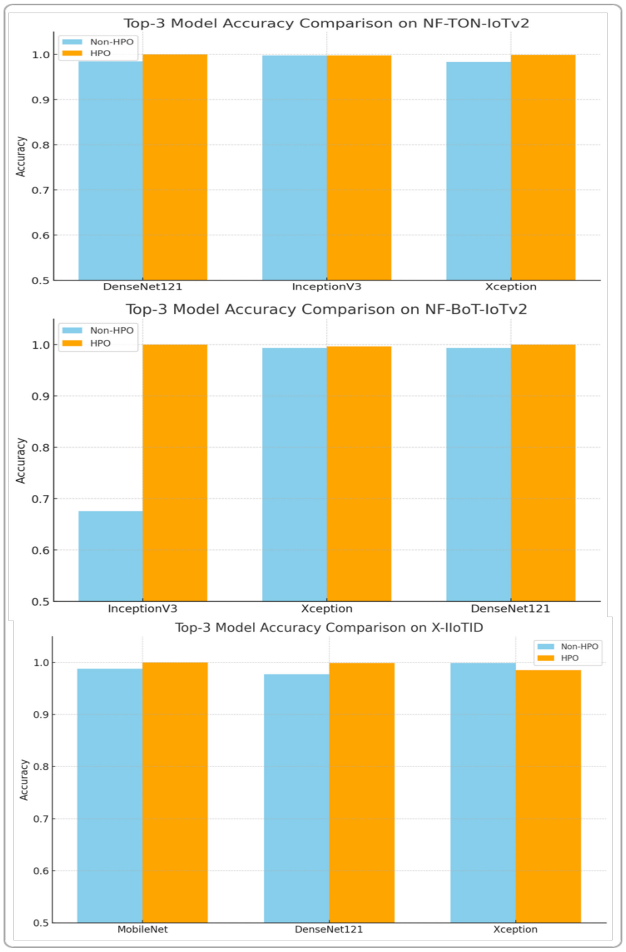

Fig. 8. Performance of the top 3 models with Non-HPO and HPO.

show the optimized accuracy for both training and validation approaches converged, while the loss of both training and validation approaches minimal value after a range of epochs. MobileNetv2's validation accuracy curve starts to converge around the 15th epoch, while the validation loss approaches the training loss by the 22nd epoch. The EfficientB0 model, like the results of the preceding dataset, differs from the other five models in that, although training accuracy grows, validation accuracy fluctuates until the optimized final epoch. The aforementioned best models are saved according to their highest validation accuracy. All of the top base learning models are saved to Kaggle's memory drive for attack predictions on unknown test datasets.

In X-IIoTID, the Xception model quickly converges, achieving near 100% training and validation accuracy by epoch 10, with training and validation loss dropping to nearly 0. InceptionV3 follows a similar trend, reaching 100% accuracy within the first few epochs and maintaining minimal loss, indicating strong generalization. MobileNet also performs exceptionally well, achieving 100% accuracy for both training and validation by epoch 6, with losses approaching 0 early in training. MobileNetV2, however, shows significant fluctuation in validation accuracy, with accuracy stabilizing around 95% after initial instability, and validation loss decreasing but not as smoothly as in the other models. DenseNet121 reaches 100% training and validation accuracy by epoch 5, with minimal loss, indicating efficient learning. EfficientNetB0 shows some variability in both validation accuracy and loss, stabilizing around 98-99% accuracy but with more fluctuation in validation loss compared to the other models.

TABLE VIII
PERFORMANCE METRICS OF HPO MODELS ON TEST DATA

Model	Performance Metrics					
	Accuracy	Precision	Recall	F1 score	CK score	MCC
NF-TON-IoTv2						
Xception	0.9990	0.9991	0.9990	0.9990	0.9990	0.9988
InceptionV3	0.9980	0.9981	0.9980	0.9980	0.9975	0.9975
MobileNet	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
MobileNetV2	0.9868	0.9882	0.9868	0.9863	0.9839	0.9840
DenseNet121	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
EfficientNetB0	0.9847	0.9839	0.9847	0.9840	0.9814	0.9814
NF-BoT-IoTv2						
Xception	0.9967	0.9968	0.9967	0.9964	0.9955	0.9955
InceptionV3	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
MobileNet	0.9951	0.9952	0.9951	0.9941	0.9933	0.9933
MobileNetV2	0.9902	0.9843	0.9902	0.9871	0.9864	0.9865
DenseNet121	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
EfficientNetB0	0.9919	0.9841	0.9919	0.9879	0.9888	0.9888
X-IIoTID						
Xception	0.9854	0.9867	0.9854	0.9831	0.9803	0.9804
InceptionV3	0.9960	0.9967	0.9960	0.9960	0.9946	0.9947
MobileNet	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
MobileNetV2	0.9774	0.9740	0.9774	0.9725	0.9696	0.9699
DenseNet121	0.9987	0.9988	0.9987	0.9987	0.9982	0.9982
EfficientNetB0	0.9854	0.9766	0.9854	0.9805	0.9804	0.9805

Table VIII provides an overview of all improved models' prediction results. Both the optimized MobileNet and DenseNet121 models performed the same in NF-TON-IoTv2. These two best-performing models correctly identify all 10 classes with 100% accuracy and achieve an impressive 100.00% attack detection accuracy. Additionally, Xception and InceptionV3 share the second position, demonstrating similar performance with accuracies of 99.90% and 99.98%, respectively. Lastly, the remaining two models, MobileNetV2 and EfficientNetB0, share the third position, attaining accuracies of 98.68% and 99.47%, respectively.

In contrast, in NF-BoT-IoTv2, the outcome differs from that of the former dataset. The top two models, InceptionV3 and DenseNet121, both get an exceptional 100.00% accuracy rate for attack detection and 100% accuracy rate for classifying all 5 classes. Moreover, Xception and MobileNet are tied for the second position, exhibiting similar performance with accuracies of 99.90% and 99.98%, respectively. Finally, MobileNetV2 and EfficientNetB0 share the third position, achieving accuracies of 99.02% and 99.19%, respectively.

In addition, in X-IIoTID dataset, MobileNet achieved perfect performance across all metrics, indicating flawless classification of all 10 classes. DenseNet121 followed closely with near-perfect scores, including 99.87% accuracy. InceptionV3 also performed strongly with 99.60% accuracy. Xception and EfficientNetB0 showed comparable results around 98.54%, while MobileNetV2 performed slightly lower at 97.74%. These results highlight MobileNet and DenseNet121 as the most effective models for X-IIoTID-based attack detection.

Since the models with HPO have been accomplished, Fig. 8 shows a comparison of the 3 selected top-performing

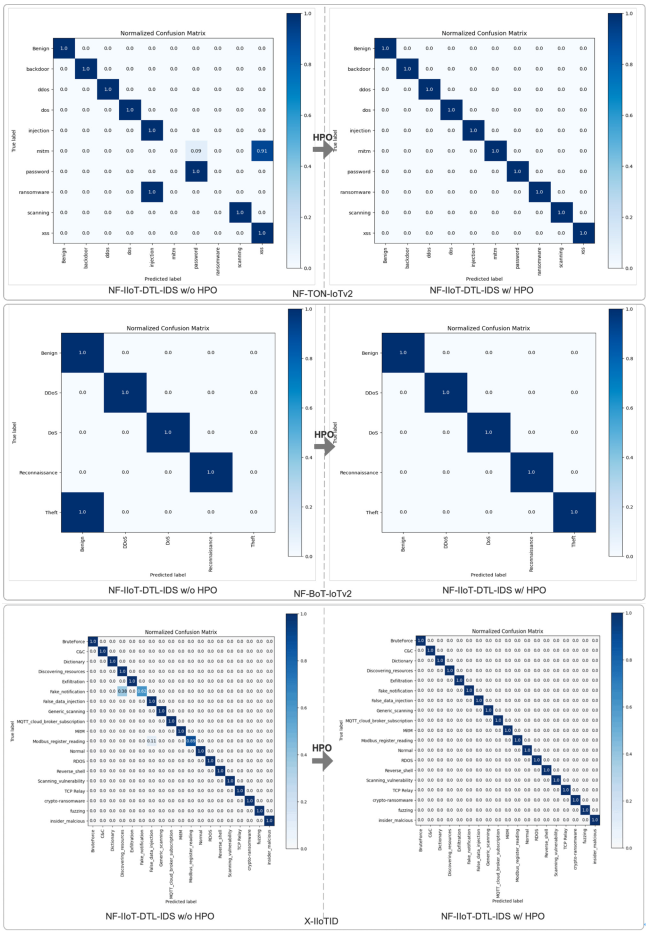

Fig. 9. Confusion matrix of NFIIoT-DTL-IDS with HPO.

models without and with HPO for three datasets, the bar charts demonstrate the improvement in accuracy for the top-3 performing models on each dataset. In Fig. 8 (NF-TON-IoTv2), DenseNet121 and MobileNet achieve perfect accuracy (100%) after HPO, with DenseNet121 improving from 98.47% and MobileNet from 94.09%. Xception also shows a significant improvement, from 98.37% to 99.90%, highlighting the effectiveness of HPO.

In addition, in Fig. 8 (NF-BoT-IoTv2), InceptionV3 sees the most dramatic accuracy increase, from 67.59% to 100%, which produce a 47.95% boost. DenseNet121 also reaches 100% accuracy post-HPO, slightly up from 99.34%. Xception improves modestly from 99.35% to 99.67%, indicating consistent gains across the board.

In Fig. 8 (X-IIoTID), MobileNet achieves 100% accuracy after HPO, rising from 98.80%, while DenseNet121 improves from 97.74% to 99.87%. Interestingly, Xception's accuracy decreases slightly post-HPO, from 99.87% to 98.54%, suggesting that HPO may not always yield improvements and highlighting the complexity of hyperparameter interactions.

In the third phase, the performance outcomes of the two soft voting ensemble models are presented and contrasted using the three datasets. Fig. 9 illustrate the normalized confusion matrices of the final ensemble model for the non-HPO (left) and HPO (right) scenarios, on NF-TON-IoTv2, NF-BoT-IoTv2, and X-IIoTID respectively. The proposed NFIIoT-DTL-IDS without Hyperparameter Optimization (HPO) accurately classifies 8 out of 10 classes in NF-TON-IoTv2, excluding MITM and Ransomware attacks, and 4 out of 5 classes in NF-BoT-IoTv2, excluding Theft attacks. However, with HPO, the proposed NFIIoT-DTL-IDS achieves precise detection of all ten classes in NF-TON-IoTv2 and all five classes in NF-BoT-IoTv2. The results indicate that the three attack classes, MITM, Ransomware, and Theft, which can be accurately identified, represent minority labels in the two datasets. In addition, for X-IIoTID, before HPO, the model performs well across most classes but struggles with the "Fake_notification" and "Modbus_register_reading" attacks, misclassifying 38% of Fake_notification as Exfiltration and 11% of Modbus_register_reading as Normal. After HPO, the model achieves perfect classification across all attack types, including previously misclassified categories, with 100% accuracy for all classes. This demonstrates a clear improvement in the model's ability to generalize and correctly classify complex attack types after optimization, particularly resolving misclassifications in the minority classes like "Fake_notification" and "Modbus_register_reading", thereby enhancing its reliability for real-world IoT intrusion detection tasks. Thus, the optimized base models with higher performance scores contribute to enhancing the performance of the final ensemble model.

Fig. 10 illustrates bar graphs comparing the key metrics of the final ensemble models for the two datasets. In Fig. 10, utilizing NF-TON-IoTv2, the attack detection accuracy of the final NFIIoT-DTL-IDS without HPO stands at 98.47%. Additionally, the other performance parameters for final precision, recall, F1-score, CK score, and MCC are 97.04%, 98.47%, 97.73%, 98.14%, and 98.16%, respectively. Conversely, the classification accuracy of the final ensemble model with HPO achieves a perfect score of 100%, along with other metrics including precision, recall, F1-score, and CK score. As depicted in Fig. 10, utilizing NF-BoT-IoTv2, the classification accuracy of the final NFIIoT-DTL-IDS without HPO is 99.35%, with other performance metrics for precision, recall, F1-score, CK score, and MCC reaching 98.73%, 99.35%, 99.03%, 99.09%, and 99.10%, respectively. Similarly, as for Fig. 10, utilizing X-IIoTID, the classification accuracy of the final NFIIoT-DTL-IDS without HPO is 99.80%, with other performance metrics for precision, recall, F1-score, CK score, and MCC reaching 98.18%, 98.80%, 98.41%, 98.39%, and 98.41%, respectively. With HPO application, all metrics achieved perfect performance at 100.00%, yielding improvements of 0.20% in accuracy, 1.82% in precision, 1.20% in recall, 1.59% in F1 score, 1.61% in Cohen's Kappa, and 1.59% in MCC. Consequently, employing the three IoT datasets, the proposed NFIIoT-DTL-IDS achieves the highest accuracy in detecting all threat classes.

B. Experimental Performance Comparison With Baseline Models

To validate the effectiveness of the proposed NFIIoT-DTL-IDS model, this section benchmarks its performance against three baseline models, including LSTM, Transformer, and

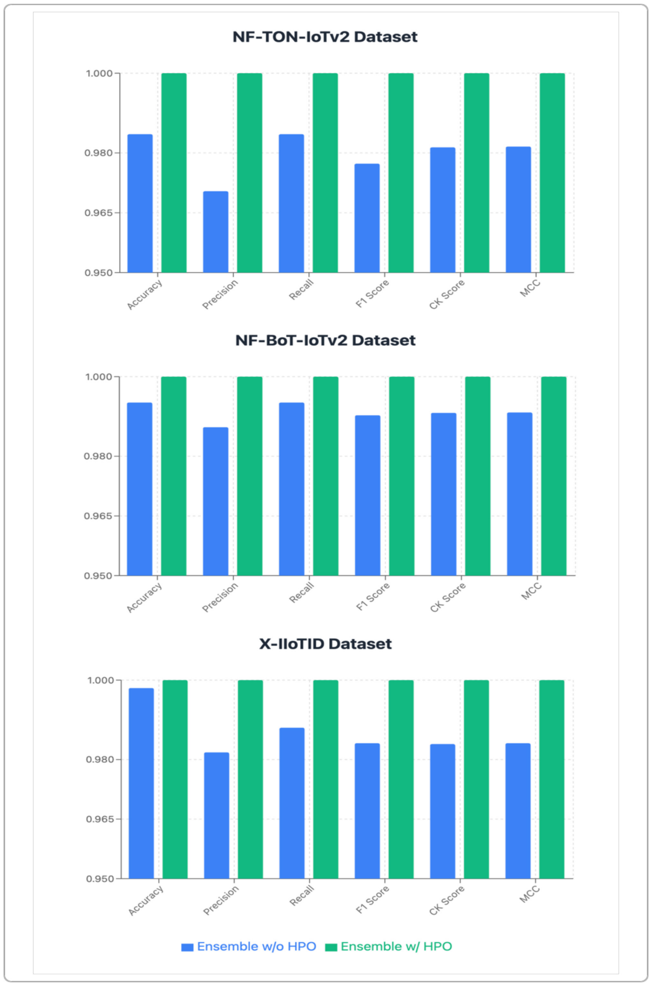

Fig. 10. Performance of the final NFIIoT-DTL-IDS models.

TABLE IX
PERFORMANCE COMPARISON WITH THE THREE BASELINE MODELS

Model	Acc	Precision	Recall	F1 Score	CK Score	MCC
NF-TON-IoTv2 Dataset						
LSTM	0.6280	0.5461	0.6280	0.5656	0.5521	0.5859
Transformer	0.9991	0.9991	0.9991	0.9991	0.9989	0.9989
3D CNN	0.9875	0.9877	0.9875	0.9874	0.9853	0.9853
NFIIoT-DTL-IDS	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
NF-BoT-IoTv2 Dataset						
LSTM	0.7298	0.6343	0.7298	0.6593	0.6486	0.7006
Transformer	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3D CNN	0.9995	0.9995	0.9995	0.9995	0.9994	0.9994
NFIIoT-DTL-IDS	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
X-IIoTID Dataset						
LSTM	0.9994	0.9995	0.9994	0.9994	0.9992	0.9992
Transformer	0.9975	0.9964	0.9975	0.9969	0.9964	0.9964
3D CNN	0.9912	0.9862	0.9912	0.9880	0.9872	0.9873
NFIIoT-DTL-IDS	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**

3D CNN, using six classification metrics, across the three benchmark datasets (NF-TON-IoTv2, NF-BoT-IoTv2, and X-IIoTID). In addition, all models were trained and evaluated over 5 independent runs to ensure the comparison results, each result reported in the tables above represents the average performance across these runs. The comparative results for each dataset are presented in Table IX.

On the NF-TON-IoTv2 dataset, the proposed NFIIoT-DTL-IDS model achieves perfect performance (100%) across all metrics, significantly surpassing LSTM by margins over 35% in all categories. Even compared to advanced models like Transformer and 3D CNN, outperforms with a 0.09–1.25% margin, demonstrating superior generalization and robustness in multi-class intrusion detection. On the NF-TON-IoTv2 dataset, the proposed model again outperforms all baseline models, maintaining flawless detection rates. While both Transformer and 3D CNN approach the upper limits of performance (especially Transformer at 100%), LSTM severely underperforms with scores below 73%, showing that traditional sequence models struggle with the complex traffic patterns in IoT environments. While LSTM achieves remarkably high performance (above 99.9%) on X-IIoTID dataset, the proposed ensemble-based NFIIoT-DTL-IDS still maintains a marginal edge by reaching the perfect score across all metrics. Compared to Transformer and 3D CNN, the proposed model improves performance by 0.25–0.88%, which indicates more consistent and error-free attack classification.

Thus, the results reveal that LSTM models consistently exhibited lower performance, particularly on the NF-TON-IoTv2 and NF-BoT-IoTv2 datasets, reflecting their limitations in capturing complex spatial-temporal patterns in IoT traffic. In contrast, Transformer and 3D CNN models demonstrated significantly better results on structured traffic, yet still underperformed compared to the NFIIoT-DTL-IDS framework. NFIIoT-DTL-IDS achieved perfect scores (100%) across all evaluation metrics and datasets, highlighting its exceptional effectiveness in multiclass intrusion detection, as well as its strong consistency across multiple runs. These results confirm the superior generalization and detection capabilities of the proposed model in diverse and dynamic IoT intrusion scenarios.

C. Performance Compared With the Recent Works

To finally assess the efficacy of the proposed NFIIoT-DTL-IDS framework, we conducted a comprehensive evaluation in three segments focusing on accuracy scores. Firstly, we compared its performance against the latest IDS on the same datasets, including NF-ToN-IoTv2, NF-BoT-IoTv2, and X-IIoTID. Then, we compared the proposed NFIIoT-DTL-IDS framework with state-of-the-art transfer learning-based intrusion detection schemes. Despite growing interest in IoT security, there remains a significant gap in the exploration of both NetFlow-based and industrial IoT scenarios, particularly in leveraging DTL methods. Most existing studies rely on outdated datasets with limited attack diversity, restricting their relevance to current threat landscapes. Moreover, DTL approaches have been largely underutilized in these contexts, limiting the adaptability and robustness of intrusion detection systems. As a result, modern IDS can only forecast a small percentage of cyberattacks.

Thus, the proposed framework uses the most recent NetFlow and Industrial-based IoT security datasets, NF-ToN-IoTv2, NF-ToN-IoTv2, and X-IIoTID which were published after 2022, and they combine NetFlow and industrial IoT-based features with more realistic and detailed data traffic to benchmark ML/DL-based IDS methods. The proposed NFIIoT-DTL-IDS, using the three datasets, can anticipate the most common types of cyberattacks in IoT networks. The second significant problem is that previous DTL research did not look into hyperparameters tuning for model training or ensemble techniques for boosting model resilience and generalization. Incorporating the renowned evolutionary algorithm, Genetic Algorithm (GA), into our proposed hyperparameter optimization strategy, we aimed to identify an optimal set of model parameters ensuring optimal performance for each base learning model. Additionally, employing the soft voting ensemble approach, we enhanced the resilience of the final NFIIoT-DTL-IDS models. As depicted in Table X, the attack detection accuracy achieved by our framework exceeds that of conventional machine learning approaches, deep learning-based IDS, and preceding DTL approaches.

TABLE X
PERFORMANCE COMPARISON OF THE PROPOSED FRAMEWORK WITH RECENT WORKS

Ref	Methodology	Dataset	Multi/Attacks	Data to Image	HPO	Ensemble	Acc (%)
NF-ToN-IoTv2							
[30]	Extra Trees	NF-ToN-IoTv2	Yes	No	No	Yes	98.05
[44]	Stacked Multi classifier		Yes	No	No	No	93.72
[14]	Line Graph Neural Network		Yes	No	No	No	93.37
NF-BoT-IoTv2							
[30]	Extra Trees	NF-BoT-IoTv2	Yes	No	No	Yes	99.99
[44]	Stacked Multi classifier		Yes	No	No	No	99.99
[14]	Line Graph Neural Network		Yes	No	No	No	95.72
X-IIoTID							
[10]	PCA/t-SNE with DT	X-IIoTID	Yes/19	No	No	No	99.45
[46]	hybrid CNN+LSTM		Yes/19	No	No	No	99.80
[47]	Self-Attention with BiLSTM		Yes/10	No	No	No	96.45
Other data sets							
[24]	TL	AWID	Yes / 6	No	No	Yes	92
[25]	TL	RPL-IIoT dataset	No	No	No	No	85.52
[48]	DTL	ISCXTor2016 ISCXVPN 2016	Yes / 4	Yes	Yes	No	94.89
[27]	DTL	CICIDS2017 NSL-KDD	Yes	Yes	Yes	Yes	99.85 99.53
[6]	DTL	ToN_IoT	Yes / 9	Yes	No	No	87
Proposed Scheme	***NFIIoT-DTL-IDS***	***NF-ToN-IoTv2 NF-BoT-IoT-v2 X-IIoTID***	***Yes /10 5/19***	***Yes***	***Yes***	***Yes***	***100***

D. Discussion

D1. Scalability

One of the key limitations in transforming NetFlow tabular data into image representations is scalability, particularly about the preprocessing overhead and storage costs. Converting a large number of rows of tabular data into image blocks significantly increases computational costs and disk usage. As the dataset grows, this transformation process becomes more resource-intensive, requiring more memory and processing time, particularly for handling high-frequency flow-based data in real-time systems. In addition, producing fixed-size image matrices from variable-length network sessions introduces rigidity, and this transformation process may limit the model's adaptability to real-world traffic scenarios with diverse characteristics. Despite these limitations, the approach offers notable advantages in terms of computational scalability once the data is transformed. Leveraging pre-trained CNNs allows the use of transfer learning, which significantly reduces training time and data requirements. The use of GPU-accelerated batch processing optimizes inference speed, making the system suitable for near-real-time deployment [49]. In addition, the entire pipeline, from data ingestion to classification, can be automated, enabling streamlined integration into modern cloud-based or edge-based IDS [50]. Thus, while initial preprocessing may be computationally expensive, the downstream benefits in terms of efficiency and deploy-ability make the model scalable in practical applications.

D2. Interpretability

Another limitation of the proposed image transformation approach is reduced interpretability. Converting the original numerical feature vectors into pixels within an image grid causes the semantic meaning of individual features to be obscured. This loss of transparency makes it difficult for security analysts to identify which specific network attributes contribute to a given classification decision. Besides, the relationships between the original features may not be easily preserved meaningfully in the form of the spatial arrangement of the transformed image style, which makes results less intuitive compared to traditional models that operate directly on raw original features. However, this limitation opens new avenues for visual interpretability from the perspective of the deep learning paradigm. Since CNNs are embedded with techniques such as Grad-CAM and saliency maps, which show important regions of an image that contribute most to the predictions [51]. These techniques produce indirect interpretation by identifying which areas, corresponding to sets or patterns of features, are most relevant to the specific attack type. Furthermore, CNNs excel at recognizing local and spatial patterns, potentially recognizing subtle traffic behaviors or attack signatures that conventional statistical methods might miss [52]. When features are stacked or grouped logically within the image, the model can learn spatial hierarchies that align with functional or protocol-based relationships, which can provide a richer understanding of network behavior.

D3. Information loss

Another challenge of image data transformation is the potential information loss, which is critical for capturing network behavior over time. The NetFlow data, by nature, consists of sequential events, in which the timing and order can be critical in detecting outliers, anomalies, and intrusion attempts. Converting this data into fixed-size image blocks may abstract away certain temporal relationships, particularly those that span across different blocks or rely on fine-grained timestamps. This consequence could cause reduced sensitivity to the type of attacks that manifest gradually or irregularly over time. However, the proposed image transformation process also brings intrinsic advantages. By leveraging 2D or 3D CNNs, the automatic learning of spatial and cross-feature correlations can be enabled, which are difficult to model with traditional tabular analysis [53]. Moreover, block-wise transformation preserves a local temporal window (e.g., 138 flows per block in the case of the NF-TON-IoTv2 dataset), allowing the model to detect short-term patterns effectively. To address the limitations regarding long-term temporal dependencies, future work could explore hybrid models—for instance, combining CNNs for feature extraction with Transformers and their variants to retain sequence-level information. This hybrid strategy can potentially provide a more holistic understanding of temporal behaviors in network flows while retaining the advantages of deep image-based learning.

D4. Model Complexity

The proposed classification model is designed to improve how well different types of attacks can be detected in IoT networks. However, this integration results in additional computational overhead, including time and computing resources, particularly in the training and optimization processes. This complexity may limit the model's real-time capability for use in resource-constrained IoT systems at the edge or fog layers, which are often deployed in IoT contexts. Thus, in future work, the exploration of lightweight models such as SqueezeNet or EfficientNet-Lite, which are designed to operate efficiently on devices with limited processing power and memory, can be employed. In addition, pruning and knowledge distillation techniques can compress and accelerate existing models without significant performance loss.

VI. Conclusion

This study introduces NFIIoT-DTL-IDS, a robust deep transfer learning-based intrusion detection framework tailored for NetFlow and Industrial IoT environments. By employing six CNN architectures, including Xception, InceptionV3, MobileNet, MobileNetV2, DenseNet121, and EfficientNetB0, with preprocessing, image transformation, hyperparameter optimization via genetic algorithm, and soft voting ensemble, the framework achieves exceptional intrusion detection accuracy. Three benchmark datasets, including NF-TON-IoTv2, NF-BoT-IoTv2, and X-IIoTID, are covered, and the model consistently achieves 100% performance across all key metrics, demonstrating its effectiveness in detecting diverse cyber threats and advanced industrial-specific attacks, such as DDoS, ransomware, and theft in common cyberattacks and MQTT subscription and crypto-ransomware attacks in industrial IoT scenarios. In addition, the study compared the proposed method with three baseline models, such as LSTM, Transformer, and 3D CNN models, and the state-of-the-art studies. NFIIoT-DTL-IDS shows superior classification capability and generalization. Despite the promising results, challenges remain in scalability, interpretability, temporal information retention, and deployment efficiency, suggesting future directions involving lightweight hybrid models and broader dataset integration. In summary, this proposed framework signifies a meaningful advancement toward securing Industry 5.0 and intelligent network management in next-generation IoT ecosystems.

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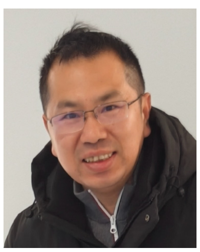

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